# Scoreboard

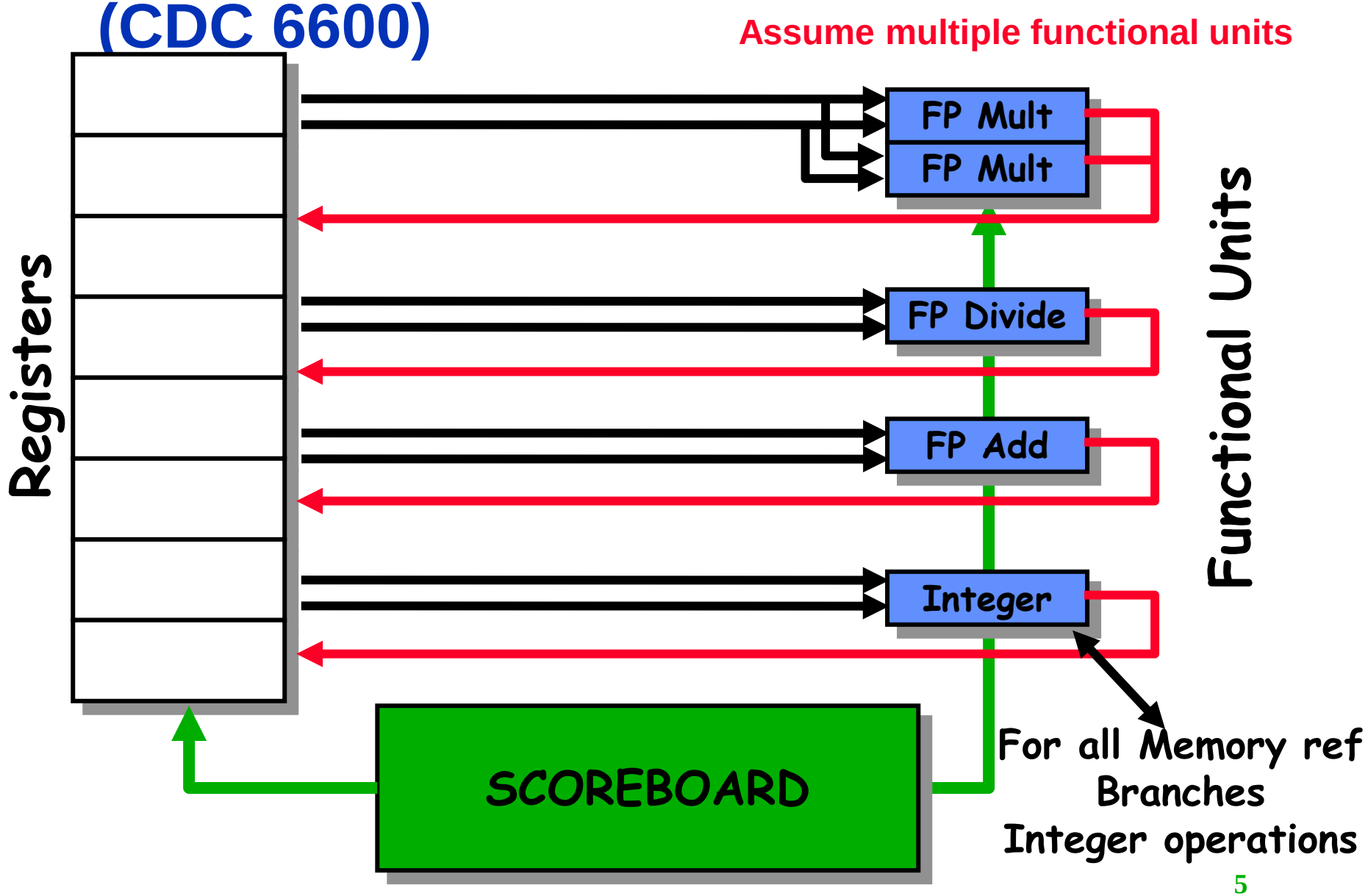
# Scoreboard: a bookkeeping technique

- **Out-of-order execution divides ID stage:**
  1. **Issue—decode instructions, check for structural hazards**
  2. **Read operands—wait until no data hazards, then read operands**
- **Scoreboards date to CDC6600 in 1963**
- **Instructions execute whenever not dependent on previous instructions and no hazards.**
- **CDC 6600: In order issue, out-of-order execution, out-of-order commit (or completion)**
  - **No forwarding!**

# Scoreboard Implications

- Out-of-order completion => WAR, WAW hazards
- Solutions for WAR:
  - Stall writeback until registers have been read
  - Read registers only during Read Operands stage
- Solution for WAW:
  - Detect hazard and stall issue of new instruction until other instruction completes
- No register renaming!
- Need to have multiple instructions in execution phase => multiple execution units or pipelined execution units
- Scoreboard keeps track of dependencies between instructions that have already issued.
- Scoreboard replaces ID, EX, WB with 4 stages

# Scoreboard Architecture (CDC 6600)



# Four Stages of Scoreboard Control

- **Issue—decode instructions & check for structural hazards (ID1)**
  - **Instructions issued in program order (for hazard checking)**
  - **Don't issue if structural hazard**
  - **Don't issue if instruction is output dependent on any previously issued but uncompleted instruction (no WAW hazards)**
- **Read operands—wait until no data hazards, then read operands (ID2)**
  - **All real dependencies (RAW hazards) resolved in this stage, since we wait for instructions to write back data.**
  - **No forwarding of data in this model**

# Four Stages of Scoreboard Control

- **Execution—operate on operands (EX)**
  - The functional unit begins execution upon receiving operands. When the result is ready, it notifies the scoreboard that it has completed execution.
- **Write result—finish execution (WB)**
  - Stall until no WAR hazards with previous instructions:

Example:

```
DIVD   F0,F2,F4
ADDD   F10,F0,F8
SUBD   F8,F8,F14
```

CDC 6600 scoreboard would stall SUBD until ADDD reads operands

# Three Parts of the Scoreboard

- **Instruction status:**
  **Which of 4 steps the instruction is in**

- **Functional unit status:—Indicates the state of the functional unit (FU). 9 fields for each functional unit**

  | | |
  |---|---|
  | **Busy:** | **Indicates whether the unit is busy or not** |
  | **Op:** | **Operation to perform in the unit (e.g., + or –)** |
  | **Fi:** | **Destination register** |
  | **Fj,Fk:** | **Source-register numbers** |
  | **Qj,Qk:** | **Functional units producing source registers Fj, Fk** |
  | **Rj,Rk:** | **Flags indicating when Fj, Fk are ready** |

- **Register result status—Indicates which functional unit will write each register, if one exists. Blank when no pending instructions will write that register**

# Scoreboard Example

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | | | | |
| LD | F2 | 45+ | R3 | | | | |
| MULTD | F0 | F2 | F4 | | | | |
| SUBD | F8 | F6 | F2 | | | | |
| DIVD | F10 | F0 | F6 | | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|
| Integer | No | | | | | | | | |
| Mult1 | No | | | | | | | | |
| Mult2 | No | | | | | | | | |
| Add | No | | | | | | | | |
| Divide | No | | | | | | | | |

*Register result status:*

| Clock | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|
| *FU* | | | | | | | | | |

# Detailed Scoreboard Pipeline Control

| Instruction status | Wait until | Bookkeeping |
|---|---|---|
| **Issue** | Not busy (FU) and not Result(D) | Busy(FU)← yes; Op(FU)← op; Fi(FU)← `D'; Fj(FU)← `S1'; Fk(FU)← `S2'; Qj← Result('S1'); Qk← Result(`S2'); Rj← not Qj; Rk← not Qk; Result('D')← FU; |
| **Read operands** | Rj and Rk | Rj← No; Rk← No; Qj ← 0; Qk ← 0 |
| **Execution complete** | Functional unit done | |
| **Write result** | ∀f((Fj(f)≠Fi(FU) or Rj(f)=No) & (Fk(f)≠Fi(FU) or Rk( f )=No)) | ∀f(if Qj(f)=FU then Rj(f)← Yes); ∀f(if Qk(f)=FU then Rk(f)← Yes); Result(Fi(FU))← 0; Busy(FU)← No |

# Scoreboard Example: Cycle 1

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | | | |
| LD | F2 | 45+ | R3 | | | | |
| MULTD | F0 | F2 | F4 | | | | |
| SUBD | F8 | F6 | F2 | | | | |
| DIVD | F10 | F0 | F6 | | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | Yes | Load | F6 | | R2 | | | Yes | Yes |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| | Divide | No | | | | | | | | |

*Register result status:*

Clock **1**

| | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|
| *FU* | | | | Integer | | | | | |

# Scoreboard Example: Cycle 2

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | | |
| LD | F2 | 45+ | R3 | | | | |
| MULTD | F0 | F2 | F4 | | | | |
| SUBD | F8 | F6 | F2 | | | | |
| DIVD | F10 | F0 | F6 | | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | Yes | Load | F6 | | R2 | | | No | No |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| | Divide | No | | | | | | | | |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **2** | *FU* | | | | Integer | | | | | |

- **Issue 2nd LD?**

# Scoreboard Example: Cycle 3

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | |
| LD | F2 | 45+ | R3 | | | | |
| MULTD | F0 | F2 | F4 | | | | |
| SUBD | F8 | F6 | F2 | | | | |
| DIVD | F10 | F0 | F6 | | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | Yes | Load | F6 | | R2 | | | No | No |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| | Divide | No | | | | | | | | |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **3** | *FU* | | | | Integer | | | | | |

- **Issue MULT?**

# Scoreboard Example: Cycle 4

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | | | | |
| MULTD | F0 | F2 | F4 | | | | |
| SUBD | F8 | F6 | F2 | | | | |
| DIVD | F10 | F0 | F6 | | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| | Divide | No | | | | | | | | |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **4** | *FU* | | | | 0 | | | | | |

# Scoreboard Example: Cycle 5

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | | | |
| MULTD | F0 | F2 | F4 | | | | |
| SUBD | F8 | F6 | F2 | | | | |
| DIVD | F10 | F0 | F6 | | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|
| Integer | Yes | Load | F2 | | R3 | | | Yes | Yes |
| Mult1 | No | | | | | | | | |
| Mult2 | No | | | | | | | | |
| Add | No | | | | | | | | |
| Divide | No | | | | | | | | |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **5** | *FU* | | Integer | | | | | | | |

# Scoreboard Example: Cycle 6

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | | |
| MULTD | F0 | F2 | F4 | 6 | | | |
| SUBD | F8 | F6 | F2 | | | | |
| DIVD | F10 | F0 | F6 | | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | Yes | Load | F2 | | R3 | | | No | No |
| | Mult1 | Yes | Mult | F0 | F2 | F4 | Integer | 0 | No | Yes |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| | Divide | No | | | | | | | | |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **6** | *FU* | Mult1 | Integer | | | | | | | |

# Scoreboard Example: Cycle 7

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | |
| MULTD | F0 | F2 | F4 | 6 | | | |
| SUBD | F8 | F6 | F2 | 7 | | | |
| DIVD | F10 | F0 | F6 | | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | Yes | Load | F2 | | R3 | | | No | No |
| | Mult1 | Yes | Mult | F0 | F2 | F4 | Integer | | No | Yes |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Sub | F8 | F6 | F2 | | Integer | Yes | No |
| | Divide | No | | | | | | | | |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | *FU* | Mult1 | Integer | | | Add | | | | |

- **Read multiply operands?**

# Scoreboard Example: Cycle 8a
## (First half of clock cycle)

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | |
| MULTD | F0 | F2 | F4 | 6 | | | |
| SUBD | F8 | F6 | F2 | 7 | | | |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | Yes | Load | F2 | | R3 | | | No | No |
| | Mult1 | Yes | Mult | F0 | F2 | F4 | Integer | | No | Yes |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Sub | F8 | F6 | F2 | | Integer | Yes | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **8** | *FU* | Mult1 | Integer | | | Add | Divide | | | |

# Scoreboard Example: Cycle 8b
## (Second half of clock cycle)

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | | | |
| SUBD | F8 | F6 | F2 | 7 | | | |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| | Mult1 | Yes | Mult | F0 | F2 | F4 | | | Yes | Yes |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Sub | F8 | F6 | F2 | | | Yes | Yes |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **8** | *FU* | Mult1 | | | | Add | Divide | | | |

# Scoreboard Example: Cycle 9

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | | |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

Note Remaining

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 10 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| 2 | Add | Yes | Sub | F8 | F6 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **9** | *FU* | Mult1 | | | | Add | Divide | | | |

- **Read operands for MULT & SUB?**
- **Issue ADDD?**

# Scoreboard Example: Cycle 10

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | | |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 9 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| 1 | Add | Yes | Sub | F8 | F6 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **10** | *FU* | Mult1 | | | | Add | Divide | | | |

# Scoreboard Example: Cycle 11

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 8 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| 0 | Add | Yes | Sub | F8 | F6 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **11** | *FU* | Mult1 | | | | Add | Divide | | | |

# Scoreboard Example: Cycle 12

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 7 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **12** | *FU* | Mult1 | | | | | Divide | | | |

- **Read operands for DIVD?**

# Scoreboard Example: Cycle 13

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | 13 | | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 6 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Add | F6 | F8 | F2 | | | Yes | Yes |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **13** | *FU* | Mult1 | | | Add | | Divide | | | |

# Scoreboard Example: Cycle 14

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 5 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| 2 | Add | Yes | Add | F6 | F8 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **14** | *FU* | Mult1 | | | Add | | Divide | | | |

# Scoreboard Example: Cycle 15

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 4 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| 1 | Add | Yes | Add | F6 | F8 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **15** | *FU* | Mult1 | | | Add | | Divide | | | |

# Scoreboard Example: Cycle 16

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 3 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| 0 | Add | Yes | Add | F6 | F8 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **16** | *FU* | Mult1 | | | Add | | Divide | | | |

# Scoreboard Example: Cycle 17

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | |

WAR Hazard!

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 2 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Add | F6 | F8 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **17** | *FU* | Mult1 | | | Add | | Divide | | | |

- **Why not write result of ADDD???**

# Scoreboard Example: Cycle 18

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 1 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Add | F6 | F8 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **18** | *FU* | Mult1 | | | Add | | Divide | | | |

# Scoreboard Example: Cycle 19

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | 19 | |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| 0 | Mult1 | Yes | Mult | F0 | F2 | F4 | | | No | No |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Add | F6 | F8 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | Mult1 | | No | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **19** | *FU* | Mult1 | | | Add | | Divide | | | |

# Scoreboard Example: Cycle 20

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | 19 | 20 |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Add | F6 | F8 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | | | Yes | Yes |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **20** | *FU* | | | | Add | | Divide | | | |

# Scoreboard Example: Cycle 21

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | 19 | 20 |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | 21 | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | Yes | Add | F6 | F8 | F2 | | | No | No |
| | Divide | Yes | Div | F10 | F0 | F6 | | | No | No |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **21** | *FU* | | | | Add | | Divide | | | |

- **WAR Hazard is now gone...**

# Scoreboard Example: Cycle 22

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | 19 | 20 |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | 21 | | |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | 22 |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| 39 | Divide | Yes | Div | F10 | F0 | F6 | | | No | No |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **22** | *FU* | | | | | | Divide | | | |

# Faster than light computation (skip a couple of cycles)

# Scoreboard Example: Cycle 61

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | 19 | 20 |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | 21 | 61 | |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | 22 |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| 0 | Divide | Yes | Div | F10 | F0 | F6 | | | No | No |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | *...* | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **61** | *FU* | | | | | | Divide | | | |

# Scoreboard Example: Cycle 62

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | 19 | 20 |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | 21 | 61 | 62 |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | 22 |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| | Divide | No | | | | | | | | |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **62** | *FU* | | | | | | | | | |

# Review: Scoreboard Example: Cycle 62

*Instruction status:*

| Instruction | | *j* | *k* | *Issue* | *Read Oper* | *Exec Comp* | *Write Result* |
|---|---|---|---|---|---|---|---|
| LD | F6 | 34+ | R2 | 1 | 2 | 3 | 4 |
| LD | F2 | 45+ | R3 | 5 | 6 | 7 | 8 |
| MULTD | F0 | F2 | F4 | 6 | 9 | 19 | 20 |
| SUBD | F8 | F6 | F2 | 7 | 9 | 11 | 12 |
| DIVD | F10 | F0 | F6 | 8 | 21 | 61 | 62 |
| ADDD | F6 | F8 | F2 | 13 | 14 | 16 | 22 |

*Functional unit status:*

| *Time* | *Name* | *Busy* | *Op* | *dest Fi* | *S1 Fj* | *S2 Fk* | *FU Qj* | *FU Qk* | *Fj? Rj* | *Fk? Rk* |
|---|---|---|---|---|---|---|---|---|---|---|
| | Integer | No | | | | | | | | |
| | Mult1 | No | | | | | | | | |
| | Mult2 | No | | | | | | | | |
| | Add | No | | | | | | | | |
| | Divide | No | | | | | | | | |

*Register result status:*

| Clock | | *F0* | *F2* | *F4* | *F6* | *F8* | *F10* | *F12* | ... | *F30* |
|---|---|---|---|---|---|---|---|---|---|---|
| **62** | *FU* | | | | | | | | | |

- **In-order issue; out-of-order execute & commit**

# Limitations of CDC 6600 Scoreboard

- **No forwarding hardware**
- **Limited to the number of scoreboard entries**
- **Limited to the number of functional units (structural hazards), especially integer/load store units**
- **Do not issue on structural hazards**
- **Wait for WAR hazards**
- **Prevent WAW hazards**

# Summary

- **Increasingly powerful (and complex) dynamic mechanism for detecting and resolving hazards**
  - **In-order pipeline, in-order op-fetch with register reservations, in-order issue with scoreboard**
  - **Allow later instructions to proceed around ones that are stalled**
  - **Facilitate multiple issue**

- **Compiler techniques make it easier for HW to find the ILP**
  - **Reduces the impact of more sophisticated organization**
  - **Requires a larger architected namespace**
  - **Easier for more structured code**